

Crack IIT JAM Smarter

AI Tools, Strategies & Techniques

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JOINT ADMISSION TEST FOR MASTERS

JAM 2027

A Graduate Student's Guide to India's Gateway Exam
for M.Sc., Dual-Degree & Integrated Ph.D. Programs at the IITs

EXAM DATE

14 Feb 2027

ORGANIZING INSTITUTE

IIT Kharagpur

TEST PAPERS

7 Disciplines

CITIES

~114 across India

What is JAM?

JAM is the national-level Computer Based Test that IIT Kharagpur is organizing for admission to postgraduate science programs at IITs and partner institutes across India for the 2027–28 academic year. It has served as the benchmark entrance exam for science education for over two decades.



Computer Based Test (CBT)

MCQ, MSQ and Numerical Answer Type questions



Open to all nationals

Indian and foreign candidates, no age restriction



One or two test papers

As long as they don't fall in the same session



~114 exam cities

Conducted across India in a single day, two sessions

Where a JAM Score Can Take You

M.Sc.

M.Sc. (Tech.)

M.S. (Research)

**M.Sc.–M.Tech.
Dual Degree**

Joint M.Sc.–Ph.D.

M.Sc.–Ph.D. Dual Degree

Integrated Ph.D.

JAM scores feed admission across 23 IITs (~3,000 seats) plus result-sharing institutes such as IISc, NITs, IISERs, DIAT, IIST Shibpur and more (~2,000 additional seats) — no separate written test or interview required.

Seven Test Papers



Highlighted: Mathematics (MA) — the paper covered in this guide's syllabus section.

Candidates may appear for ONE or TWO test papers, provided they are not scheduled in the same session.

Exam Schedule

14

FEBRUARY
2027
SUNDAY

Session	Time	Test Papers
Forenoon (FN)	9:30 am – 12:30 pm	CY, GG, MA
Afternoon (AN)	2:30 pm – 5:30 pm	BT, EN, MS, PH

Duration: 3 hours per session • A candidate may attempt two papers only if scheduled in different sessions

Compensatory Time for PwD Candidates

PwD candidates are eligible for one hour of compensatory time per paper (ending 1:30 pm for FN, 6:30 pm for AN), and may opt for scribe services by indicating this during registration.

Who Can Apply?

1 Completed or final-year undergraduates

Candidates who have completed their undergraduate degree, or are appearing in the final examination of their qualifying degree in 2027, are eligible to appear.

2 Degree must be complete before admission

The undergraduate program should be fully COMPLETED before the date of admission at the respective Admitting Institute.

3 Proof of eligibility required

Proof of passing the qualifying degree, meeting the Admitting Institute's specified eligibility, must be submitted within its given timeline.

4 No age restriction, open to all nationals

Indian and foreign nationals may apply; foreign nationals must also satisfy the admitting institute's rules for international students.

Exam Pattern: Three Question Types

SECTION A

MCQ

Multiple Choice Question

- Only one correct option out of four
- Negative marking applies
- $-1/3$ mark (1-mark Q)
- $-2/3$ mark (2-mark Q)

SECTION B

MSQ

Multiple Select Question

- One or more correct options
- No negative marking
- No partial marking either

SECTION C

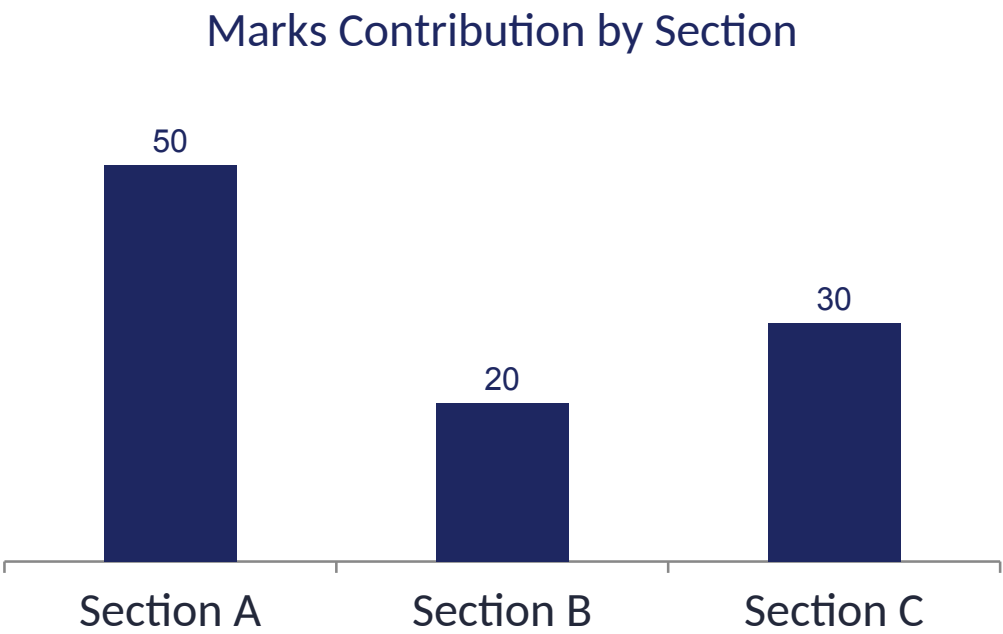
NAT

Numerical Answer Type

- Answer entered via virtual keypad
- No answer choices shown
- No negative marking

Structure & Marking Scheme

Section	Questions	Marks Each	Total
A: MCQ (Q1–10)	10	1	10
A: MCQ (Q11–30)	20	2	40
B: MSQ (Q31–40)	10	2	20
C: NAT (Q41–50)	10	1	10
C: NAT (Q51–60)	10	2	20



All sections are compulsory. Unattempted questions score zero. A virtual on-screen calculator is provided; personal calculators are not allowed.

Application Fees

Female / SC / ST / PwD

One Test Paper

₹1,000

Two Test Papers

₹1,350

Non-refundable under any circumstances

All Other Candidates

One Test Paper

₹2,000

Two Test Papers

₹2,700

Non-refundable under any circumstances

Correcting category, gender, exam city or date of birth costs an extra ₹300. Adding or changing a test paper on an already-submitted form also costs ₹300, and must be done before 10 November 2027.

Where You Could Be Admitted

ADMITTING INSTITUTES

23

IITs

≈ 3,000 postgraduate seats. JAM score alone decides admission — no extra interview or test.

RESULT SHARING INSTITUTES

2,000+

seats

At IISc, NITs, CFTIs, IISER Pune & Bhopal, DIAT, IIST Shibpur, IIPE, JNCASR, SLIET and more.

Admission is based on All-India Rank (AIR) in each paper, subject to eligibility requirements, reservation policy, and seat availability.

Mathematics (MA) Syllabus Snapshot

1. Real Analysis

- Sequences & series, Cauchy sequences, Bolzano–Weierstrass theorem
- Convergence tests; power series & term-wise differentiation/integration
- Continuity, differentiation, Rolle's & mean value theorems, Taylor's series
- Riemann integration & the fundamental theorem of calculus

2. Multivariable Calculus & Differential Equations

- Limits, continuity, partial & total derivatives, maxima/minima
- Double & triple integrals; areas and volumes
- Bernoulli's, exact, homogeneous & linear 2nd-order ODEs; Cauchy-Euler equation

3. Linear Algebra & Algebra

- Permutations, combinations & the binomial theorem
- Matrices: rank, nullity, inverse, determinant, eigenvalues/eigenvectors
- Vector spaces, linear transformations, rank-nullity theorem
- Groups: cyclic, abelian, permutation groups, homomorphisms

IIT JAM Mathematics – Topic-wise Previous Year Questions

Topic	2020	2021	2022	2023	2024	2025	2026	Average
Real Analysis	33	32	20	34	36	34	36	31.57
Modern Algebra (Group Theory)	10	17	11	13	16	13	16	13.71
Linear Algebra	16	19	23	21	17	21	17	19.57
ODE	13	11	16	10	14	10	14	12.57
Integral Calculus	12	7	17	10	13	10	13	11.71
Functions of several variables	10	8	13	12	4	12	4	9.00
Total	94	94	100	100	100	100	100	98.29

IIT JAM Mathematics Previous Year Cut-Offs

Cut-Off Marks (Out of 100)

Year	General (GEN)	OBC-NCL / EWS	SC / ST / PwD
2026	12.65	11.38	6.32
2025	19.90	17.91	9.95
2024	11.22	10.09	5.61
2023	24.56	22.10	12.28
2022	22.91	20.62	11.46
2021	24.69	22.22	12.34

Prompting Techniques That Actually Help

Technique	Why it works for JAM prep
Be specific about level	“Explain at 10+2 / undergraduate level” keeps answers matched to JAM’s syllabus depth, not research-level detail
Ask for hints before solutions	Preserves the struggle that builds problem-solving speed for a timed CBT
Request step-by-step working	Mirrors how you must reconstruct NAT answers without a fixed option to fall back on
Ask “why is this option wrong?”	Trains MCQ elimination skills directly, not just the correct answer
Give it your actual mistakes	Specific, personalized follow-up beats generic “explain topic X” every time

Where AI Helps – and Where It Doesn't

Use it for

- ▶ Explaining concepts and worked examples
- ▶ Drafting a study plan and revision schedule
- ▶ Turning notes into flashcards or summaries
- ▶ Spotting patterns across mock-test mistakes

Don't rely on it for

- ▶ Timed practice – the real exam has **no AI access**, so simulate that
- ▶ Final answer verification – **cross-check** against your textbook or standard reference
- ▶ Numerical computation you should be able to do by hand under time pressure
- ▶ Replacing full-length mock tests – analytics support mocks, they don't substitute for them

Exam-Day Section Strategy Recap

Section	Tactic
A (MCQ)	Negative marking means only attempt when you can eliminate at least 2 of 4 options ; skip pure guesses
B (MSQ)	No penalty for wrong selections – attempt every question ; mark options you're confident about even if unsure of others
C (NAT)	No negative marking and no options to eliminate from – attempt all , and use the on-screen virtual calculator to sanity-check arithmetic

Practice this exact triage – skip, flag, attempt – in every AI-analyzed mock so it's automatic by exam day, 14 February 2027.

KEY TAKEAWAYS

Your Next Steps Toward JAM 2027

1

Confirm eligibility & pick your paper(s)

Check ERs/MEQs of your target program and choose up to two non-overlapping test papers.

2

Register on JOAPS

Complete the online application, upload photograph & signature, and pay the fee before the deadline.

3

Practice the pattern

Take the official mock test and review previous years' papers — focus on MCQ, MSQ and NAT formats.

4

Sit for the exam — 14 Feb 2027

Arrive early, carry your admit card, and remember: no personal calculators, only the on-screen one.

jam.iitkgp.ac.in

Prepare smarter. Practice sharper.

All the best for JAM 2027 – 14 February 2027
